

CPSC 304 Project Cover Page

Milestone #: 1

Date: 02/26/26

Group Number: 36

Name	Student Number	CS Alias (Userid)	Preferred E-mail Address
Julian Wright	84895358	w6i9k	jwrigh13@student.ubc.ca
Dorson Tang	94262474	d5v6x	dtang08@student.ubc.ca
Krishna Garcha	40548323	g8a8b	garchakri@gmail.com

By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

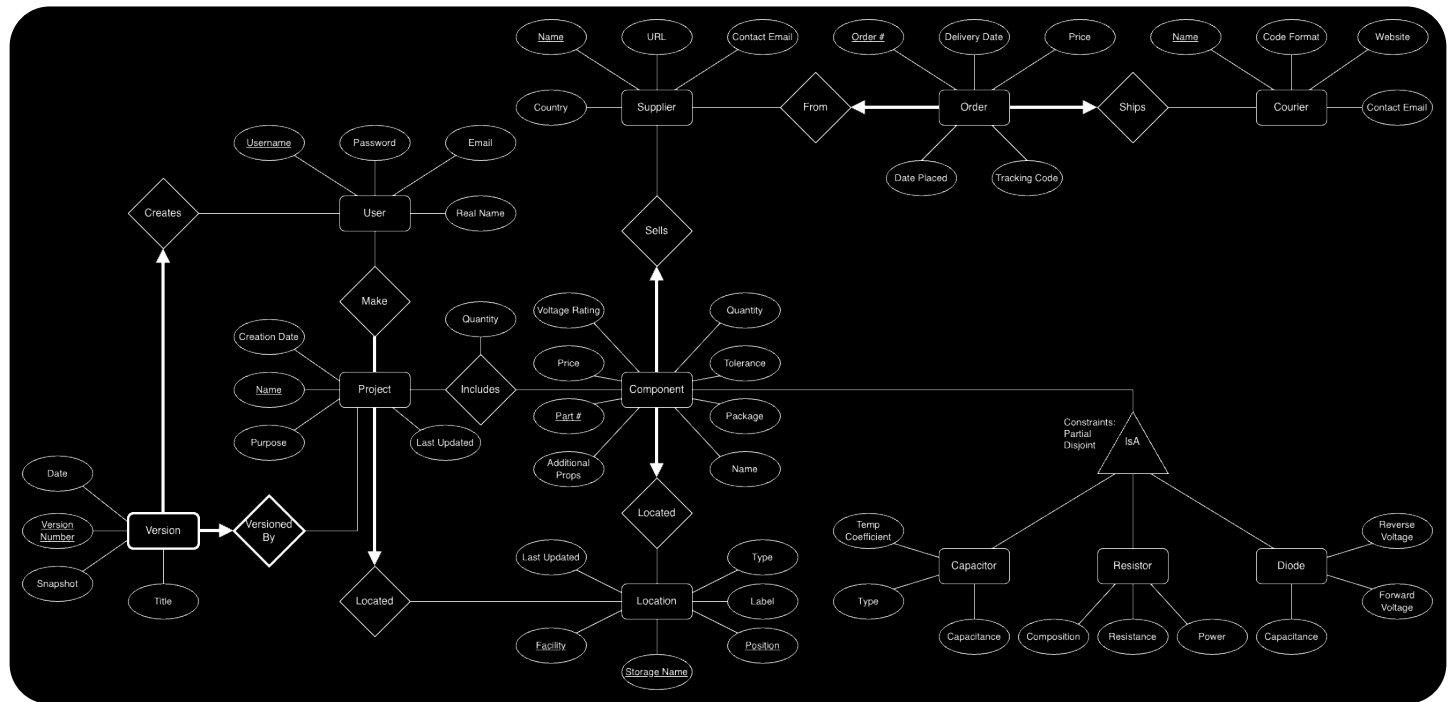
Project Description

The domain of our project is electronics project management. The idea is to allow users to create projects and add electronic components (such as resistors, capacitors, etc.) for organizational/tracking purposes. They can also choose suppliers for the components and track orders from those suppliers. Users can add locations (e.g., drawers, component bins, etc.) to both their components and projects to be able to find them better, something especially important for the components as they can get lost easily.

Technology

We plan to use TypeScript with React and PostgreSQL for our project.

ER Diagram



Schema

Component(part_num: string, price: float, name: string, package: string, tolerance: float, quantity: int, voltage_rating: float, additional: json, storage_name: string, position: string, facility: string, supplier_name: string)

Capacitor(temp_coeff: string, type: string, capacitance: float, part_num)

Resistor(composition: string, resistance: float, power: string, part_num)

Diode(capacitance: float, vforward: float, vreverse: float, part_num)

Project(name: string, created_at: timestamp, purpose: string, last_updated: timestamp, storage_name: string, position: string, facility: string)

Version(project_name: string, version_number: string, title: string, date: timestamp, snapshot: jsonb, created_by: string)

User(username: string, password: string, email: string, real_name: string)

Order(order_number: string, price: float, tracking_code: string, date_placed: string, delivery_date: string, supplier: string, courier: string)

Supplier(name: string, url: string, contact_email: string, country: string)

Courier(name: string, code_format: string, website: string, contact_email: string)

Location(storage_name: string, position: string, facility: string, type: string, label: string, last_updated: timestamp)

Includes(project_name: string, component_part_num: string, quantity: int)

Makes(project_name: string, username: string)

Functional Dependencies

Component

- part_num -> price, name, package, tolerance, quantity, voltage_rating, additional, storage_name, position, facility, supplier_name

Capacitor

- part_num -> temp_coeff, type, capacitance

Resistor

- part_num -> composition, resistance, power

Diode

- part_num -> capacitance, vforward, vreverse

Project

- name -> created_at, purpose, last_updated, storage_name, position, facility

Version

- name, version_number -> title, data, snapshot, created_by
- snapshot -> title

User

- username -> password, email, real_name
- email -> username
- email -> real_name

Order

- order_number -> price, tracking_code, date_placed, delivery_date, supplier_name, courier_name
- tracking_code -> delivery_date

Supplier

- name -> url, contact_email, country

Courier

- name -> code_format, website, contact_email

Location

- storage_name, position, facility -> type, label, last_updated

Includes

- project_name, component_part_num -> quantity

Makes

N/A

Normalization

Component(part_num: string, price: float, name: string, package: string, tolerance: float, quantity: int, voltage_rating: float, additional: json, **storage_name: string, position: string, facility: string, supplier_name: string**)

Capacitor(temp_coeff: string, type: string, capacitance: float, **part_num: string**)

Resistor(composition: string, resistance: float, power: string, **part_num: string**)

Diode(capacitance: float, vforward: float, vreverse: float, **part_num: string**)

Project(name: string, created_at: timestamp, purpose: string, last_updated: timestamp, **storage_name: string, position: string, facility: string**)

Version(**project_name: string, version_number: string**, title: string, date: timestamp, snapshot: jsonb, **created_by: string**)

Version1(snapshot: jsonb, title: string)

Version2(**project_name: string, version_number: string, created_by: string**, date: timestamp, snapshot: jsonb)

User(username: string, password: string, email: string, real_name: string)

User1(email: string, username: string)

User2(password: string, email: string, real_name: string)

User3(email: string, real_name: string)

User4(password: string, email: string)

Order(order_number: string, price: float, tracking_code: string, date_placed: string, delivery_date: string, **supplier_name: string, courier_name: string**)

Order1(tracking_code: string, delivery_date: string)

Order2(order_number: string, price: float, tracking_code: string, date_placed: string, **supplier_name: string, courier_name: string**)

Supplier(name: string, url: string, contact_email: string, country: string)

Courier(name: string, code_format: string, website: string, contact_email: string)

Location(storage_name: string, position: string, facility: string, type: string, label: string, last_updated: timestamp)

Includes(**project_name: string, component_part_num: string**, quantity: int)

Makes(**project_name: string, username: string**)

SQL DDL

```
-----
CREATE TABLE Component (
    part_num    VARCHAR PRIMARY KEY,
    price       DECIMAL(10,2),
    name        VARCHAR NOT NULL,
    package     VARCHAR,
    tolerance   DECIMAL(6,6),
    quantity    INT DEFAULT 0,
    voltage_rating DECIMAL(8,4),
    additional  JSONB,

    storage_name VARCHAR,
    position     VARCHAR,
    facility     VARCHAR,

    supplier_name VARCHAR NOT NULL,

    FOREIGN KEY (storage_name, position, facility)
        REFERENCES Location(storage_name, position, facility)
        ON DELETE SET NULL,

    FOREIGN KEY (supplier_name)
        REFERENCES Supplier(name)
        ON DELETE NO ACTION
);

CREATE TABLE Capacitor (
    temp_coeff  VARCHAR,
    type        VARCHAR,
    capacitance DECIMAL(8,4) NOT NULL,

    part_num    VARCHAR PRIMARY KEY,

    FOREIGN KEY (part_num)
        REFERENCES Component(part_num)
        ON DELETE CASCADE
);

CREATE TABLE Resistor (
    composition VARCHAR,
    resistance   DECIMAL(8,4) NOT NULL,
    power        VARCHAR,

    part_num    VARCHAR PRIMARY KEY,

    FOREIGN KEY (part_num)
        REFERENCES Component(part_num)
        ON DELETE CASCADE
);

CREATE TABLE Diode (
    capacitance DECIMAL(8,4),
    vforward    DECIMAL(8,4) NOT NULL,
    vreverse     DECIMAL(8,4) NOT NULL,

    part_num    VARCHAR PRIMARY KEY,

    FOREIGN KEY (part_num)
        REFERENCES Component(part_num)
        ON DELETE CASCADE
);

-----
CREATE TABLE User (
    username  VARCHAR PRIMARY KEY,
    password  VARCHAR NOT NULL,
    email     VARCHAR NOT NULL UNIQUE,
    real_name VARCHAR NOT NULL
);

CREATE TABLE Location (
    storage_name VARCHAR,
    position     VARCHAR,
    facility     VARCHAR,
    type         VARCHAR,
    label        VARCHAR,
    last_updated TIMESTAMP,

    PRIMARY KEY (storage_name, position, facility)
);

CREATE TABLE Supplier (
    name        VARCHAR PRIMARY KEY,
    url         VARCHAR NOT NULL,
    country     VARCHAR NOT NULL,
    contact_email VARCHAR
);
```

```
CREATE TABLE Courier (
    name        VARCHAR PRIMARY KEY,
    code_format VARCHAR,
    website     VARCHAR NOT NULL,
    contact_email VARCHAR
);

CREATE TABLE Version (
    version_number VARCHAR,
    title          VARCHAR NOT NULL,
    date           TIMESTAMP NOT NULL,
    snapshot       JSONB NOT NULL,
    created_by     VARCHAR,
    project_name   VARCHAR,

    FOREIGN KEY (created_by)
        REFERENCES User(username)
        ON DELETE SET NULL,

    FOREIGN KEY (name)
        REFERENCES Project(name)
        ON DELETE CASCADE,

    PRIMARY KEY (project_name, version_number)
);

-----
CREATE TABLE Project (
    name        VARCHAR PRIMARY KEY,
    created_at  TIMESTAMP NOT NULL,
    purpose     TEXT,
    last_updated TIMESTAMP,

    storage_name VARCHAR,
    position     VARCHAR,
    facility     VARCHAR,

    FOREIGN KEY (storage_name, position, facility)
        REFERENCES Location(storage_name, position, facility)
        ON DELETE NO ACTION
);

CREATE TABLE Order (
    order_number  VARCHAR PRIMARY KEY,
    price         DECIMAL(10,2) NOT NULL,
    tracking_code VARCHAR,
    date_placed   DATE NOT NULL,
    delivery_date DATE,

    supplier      VARCHAR NOT NULL,
    courier       VARCHAR,

    FOREIGN KEY (supplier)
        REFERENCES Supplier(name)
        ON DELETE SET NULL,

    FOREIGN KEY (courier)
        REFERENCES Courier(name)
        ON DELETE SET NULL
);

-----
CREATE TABLE Includes (
    project_name  VARCHAR,
    component_part_num VARCHAR,
    quantity      VARCHAR NOT NULL,

    PRIMARY KEY (project_name, component_part_num),

    FOREIGN KEY (project_name)
        REFERENCES Project(name)
        ON DELETE CASCADE,

    FOREIGN KEY (component_part_num)
        REFERENCES Component(part_num)
        ON DELETE CASCADE
);

CREATE TABLE Makes (
    project_name VARCHAR,
    username     VARCHAR,

    PRIMARY KEY (project_name, username),

    FOREIGN KEY (project_name)
        REFERENCES Project(name)
        ON DELETE CASCADE,

    FOREIGN KEY (username)
        REFERENCES User(username)
```

```
REFERENCES User(username)
ON DELETE CASCADE
);
```

AI Tool Use

No AI tools were used for this milestone.